TIPA Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	**Localization and Perception for Autonomous Vehicles: Radar-based Point-Cloud Map-Matching, Free Space Detection and Object Detection.**			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery ( ), Advanced Mobility (√), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Mechanical and Materials		Mechanical and Materials	
	Medium Category	Small Category	Medium Category	Small Category
	Automobile / Rail Vehicles	Safety Improvement Technologies	Automobile / Rail Vehicles	Vehicle Intelligence Technologies
Concept of Proposed Technology	For higher levels of vehicle automation, it is crucial to have accurate localization (exact positioning and pose estimation in the global or local frame), and perception (detection of key features around the automated vehicle) at all times while the vehicle automation is operational. However, each individual sensor has their own sets of strengths and weaknesses, in terms of performance characteristics and their operational domains. For instance, cameras are often impacted by poor weather, while GNSS sensors are unavailable in tunnels and other indoor environments. Therefore, a combination of different sensors is often used, to maximise the availability and robustness of the localization and perception outcomes. However, this often comes at higher costs, making it harder to scale autonomous vehicles. Due to their ability to work better than LiDAR and camera sensors during bad weather conditions, as well as the longer range, there is an increasing interest from the industry to utilize Radar sensors. Radars can also use their doppler measurements to estimate the radial velocity of objects on the fly, which is also useful for automated driving applications. However, in TNO experience, the density of information delivered by typical Radar sensors is not			

adequate to perform accurate localization and perception. Therefore, they are often used as a part of a composite sensor set, in combination with cameras and LiDARs.

The upcoming generation of 4D/imaging Radars aims to partly address this, with a much higher density of information contained within the point-cloud delivered by the Radar, but this needs further development on both the hardware and software side, to scale their application.

In this project, TNO, together with the partner Korean SMEs, will make steps to further develop, optimize and scale the application of the imaging Radars for accurate (point-cloud) localization and perception.

This will be enabled by three main functionalities:

1. Point-cloud map-matching;
2. Free-space detection; and
3. Object detection.

The end objective is to enable localization and perception applications for an automated vehicle wherein the vehicle can find its accurate position and pose, as well as identify what is around the vehicle, in terms of free space that the vehicle can maneuver into, as well as objects that it should take into account.

Point-cloud map-matching

Point-cloud map-matching is a localization technique that utilizes the point-clouds (mapping of reflections in space), generated by a LiDAR or Radar sensor, to identify the positioning of the sensor with respect to a pre-generated (high-definition) map. Usually, (variations of) one of two algorithms is utilized for this – Normal Distributions Transform (NDT), or Iterative Closest Points (ICP). As a pre-step, however, the outputs of the various sensors (e.g., multiple Radars with different headings) may need to be combined into a unified point-cloud that can provide, for instance, a 360-degree view around the ego vehicle.

TNO has significant experience with utilizing NDT-based algorithms for LiDAR and conventional Radar point-cloud map-matching. Within this project, in collaboration with a Tier-1 or Tier-2 supplier, TNO would like to extend the functionality towards the utilization of imaging Radars for point-cloud map-matching.

Additionally, this includes the development of a functionality that can combine the output of multiple Radar sensors into a combined point-cloud.

Free space detection

The information gathered from the Radar or other sensors (LiDAR/camera) will be used for identifying what free space exists around the vehicle. This includes taking into account detected barriers, poles as well as other road users. This should interface with a motion planning stack that may be supplied by an AD stack supplier.

Object detection

TNO has experience developing object detection, classification and tracking, using LiDAR and camera information. While it is possible to detect the presence of objects with a conventional Radar sensor, the classification and tracking of such objects is very challenging, owing to the limited information.

Therefore, in this step, the previous methods developed by TNO will be extended towards object classification and tracking based on the information provided by the imaging Radars. In collaboration with the SME, the Radar output can also be tuned, to enable better detection, based on the different scenarios that the vehicle can encounter. Additionally, if possible, the object detection framework will be trained on annotated data provided by the Korean SME, to tune the algorithm for the Korean/other market.

Rationale for Joint R&D Project	Given the relatively new nature of imaging Radars, and open research questions on their application as a fall-back or complementary localization/perception technique, additional R&D effort is needed to develop this functionality. This aligns with the fact that the automotive supply chain is extremely international in terms of stakeholders. Given the leading position of Korea, as well as high knowledge position of Dutch research institutes in the smart mobility and automotive domain, we believe that this collaboration would benefit the innovation chain on both sides.
Key Words	1. Automated vehicle localization. 2. Local HD map generation. 3. Radar based object detection and classification.

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	**Start of the Project**	Level (3)	**End of the Project**	Level (6)
R&D Objectives	The final goal of the project is to develop an imaging Radar based point-cloud map-matching, and free-space as well as object detection algorithms, for use in automated mobility applications.			
R&D Target Specifications	**Vehicle positioning** estimation error below 0.3m Root-Mean-Squared-Error (RMSE) • **Object detection and classification** ○ Mean Average Precision (mAP) of highway users up to 0-30m: 80-90% ○ Mean Average Precision (mAP) of highway users up to 30-60m: 60-70% ○ Mean Average Precision (mAP) of highway users up to 60-90m: 30-50%			
R&D Plan	The proposed project will follow a phased execution plan, with an incremental innovation and testing process. This will include multiple different phases: **TNO Proposal: Localization and Perception for Autonomous Vehicles: Imaging Radar-based Point-Cloud Map-Matching, Free Space Detection and Object Detection** Phase 1: Architecture specification and requirements definition → Phase 2: Imaging Radar installation, integration and configuration → Phase 3: Imaging Radar-based point-cloud map matching development → Phase 4: Free space detection functionality → Phase 5: Object detection functionality **Phase 1: Architecture specification and requirements definition (Months 0-3; Lead: TNO)** The requirements will be specified by TNO, based on the use cases foreseen by the Korean SME. This includes defining safety requirements as well as functional targets for the various use cases. Concurrently, TNO will define an architecture for the localization framework and the functionalities mentioned before. Milestone M1: Shared architecture and requirements defined and documented. Deliverable D1 (TNO): Architecture and requirements specification.			

Phase 2: Imaging Radar installation, integration and configuration (Months 0-6; Lead: SME)

The SME will develop and deliver to TNO test prototypes of their imaging Radars. This will be configured and installed on the TNO Carlab test platforms for on-road testing. Additionally, shared test environments, data sharing platforms and common interfaces in the ROS2 middleware environment will be set up. The SME will provide support in calibration and tuning of the imaging Radar.

Milestone M2: Radars delivered, configured and installed. Ready for conducting testing.

Deliverable D2 (SME): Imaging Radars delivered to TNO, configured and installed at TNO.

Phase 3: Imaging Radar-based point-cloud map matching development (Months 6-10; Lead: TNO)

TNO will extend their point-cloud map-matching framework to incorporate the imaging Radar system provided by the SME. This includes developing a function to combine the point-cloud of multiple Radars.

This will be tested on the TNO Carlab, in the Netherlands, where TNO has access to (limited stretches of) HD maps. The performance will be evaluated and quantified, based on a high-accuracy RTK-GNSS system, which would be considered to be the ground truth data.

Milestone M3: Point-cloud map matching function active, and knowledge transferred to SME.

Deliverable D3 (TNO): Point cloud map matching software module.

Phase 4: Free space detection functionality (Months 10-16; Lead: TNO)

TNO will develop a functionality to detect the free space with respect to the ego automated vehicle. This will be specified in accordance with the architecture and requirements specified in Phase 1, to ensure that this is compatible with motion planning stacks. The SME will provide continued support for the calibration of the Radar.

This will be tested in the TNO Carlab in the Netherlands, while operating in a non-actuated, ghost mode, as well as on the test/data collection platform at the SME in Korea.

Milestone M4: Free space detection functionality implemented and active and knowledge transferred to SME.

Deliverable D4 (TNO): Free space detection software module.

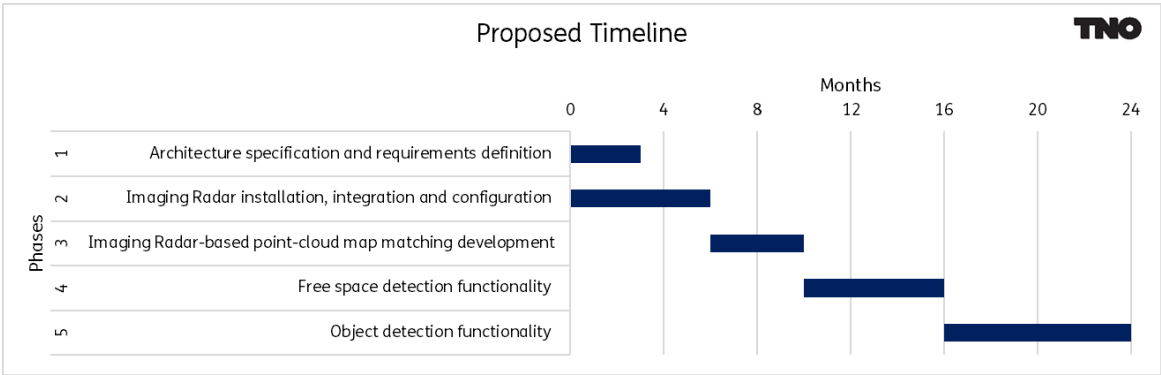
Phase 5: Object detection functionality (Months 16-24; Lead: TNO)

If possible, the SME will provide a dataset of annotated object information collected on Korean roads. This will be used to develop and train the TNO object detection framework to function using Radar information. This framework yields bounding boxes and labels for recognized objects in the surroundings of the ego vehicle.

If this data is not available, TNO will use the existing available (European/Dutch) datasets to train the object detection algorithm. Based on the source of the training dataset, this functionality would be tested in road tests in Korea/the Netherlands, and analyzed by TNO by means of benchmarking with respect to LiDAR/camera based object detection algorithms available at TNO.

Milestone M5: Object detection functionality implemented, active and benchmarked. Workshops conducted to provide knowledge to the SME on how to incorporate the outcomes in their development approach.

Deliverable D5 (TNO): Object detection software module.



4. R&R (Roles and Responsibilities)

Expected R&R of TNO	TNO (particularly the Integrated Vehicle Safety research group) will provide the primary technical expertise and lead the research activities within this program. TNO will utilize their algorithm development experience and dedicated research platforms to undertake this research. As a part of this collaboration, TNO will undertake the following activities: **Develop an imaging Radar-based point-cloud map matching and fusion algorithm** TNO has significant experience with LiDAR-based point-cloud map matching algorithms. However, within this project, this algorithm will be extended for utilizing point-cloud information coming from a (set of) imaging Radars. This fusion algorithm will fuse the information coming from multiple imaging Radars, and utilize state-of-the-art methods to provide a combined location and pose estimate of the ego vehicle. This will be tested using TNO's Carlabs in the Netherlands. The performance validation methodology for the point-cloud map matching functionality will be shared with the SME. **Develop a free-space detection algorithm** Together with the SME, TNO will specify a scalable free-space detection algorithm such that this can directly interface with a (variety of) motion planning stack(s). Based on this, TNO will develop a free-space detection algorithm, to evaluate the free space available around the ego (autonomous) vehicle. This will output an estimate of the available space that the vehicle can use to maneuver. As much as possible, this module will be agnostic of the sensor set used, to ensure that this can also be scaled to LiDAR or camera based detections. This will be tested on the TNO Carlabs in the Netherlands, or on the test platforms of the partner SME. The performance validation methodology for the free-space detection functionality will be shared with the SME. **Develop an object detection algorithm** TNO will develop and optimize an object detection algorithm, to detect and classify objects with regards to the ego vehicle. This will include a bounding box per object, as well as a label that provides a classification of the object (e.g., car, pedestrian etc.). This will be trained based on existing datasets of European roads. If a significant dataset is available, this can also be trained for Korean roads. **Knowledge transfer to Korean SMEs** One of the primary objectives of this program is to enable cross-industry and cross-country learning for both TNO and the partner SMEs. This will be facilitated through the ongoing discussions and combined (virtual) workshops, to share knowledge and outcomes together. This will also include the shaping of project outcomes together, including e.g., test plans etc., so as to maximise application and business value on both sides. Additionally, gathered data would be accessible for both sides. Finally, TNO would provide support to the SME on how the outcomes of this project can be incorporated into their validation processes.

Expected R&R of Korean SME Partner	The partnering Korean SME(s) are responsible for development of the Radars hardware, as well as for being the problem owners – setting requirements and specifications for the functionality development. The envisioned tasks for the SME are: **Overall project management** The Project Management will ensure that all tasks are executed efficiently and within the established constraints, while achieving the required goals. This includes monitoring and mitigating project risks, in so far as possible on the project level. **Requirement specification and architecture design** The SME will lead Phase 1, and would be responsible for the specification oof the requirements and the design of the functional architecture (with TNO). This is done to enable the SME to shape the outcomes in a way that enables maximum alignment with industrial partners. **Development and tuning of the imaging Radars** The SME is responsible for developing, providing and tuning the imaging Radar hardware environment, as well as providing any software needed to arrive at the agreed upon middle-ware required by TNO. **Data collection** In Phase 5, if this is desired by the SME as the training dataset, the SME is requested to provide a dataset of annotated object information from the Korean roads. **Public-road testing** In this project, TNO will follow a co-development approach. Therefore, in the Netherlands, TNO will conduct testing and validation of the algorithms developed in the project. To ensure cross-border compatibility, the Korean SME is requested to do so on Korean roads.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	**Outcomes:** This project leads to the following outputs: • Imaging-Radar based point-cloud map matching and multi-Radar fusion algorithm. • Free space detection algorithm. • Object detection algorithm. • Optimised and tested imaging Radars. **Impact:** Imaging Radars are expected to gain further prominence for autonomous vehicles as the technology matures. This project aims to provide the partner SME with a direct competitive advantage in the maturity of their technology. This development is expected to be directly applicable to the partner SME's workflow. The main objective would be to enable the SMEs to: • Demonstrate and quantify the effectiveness of their hardware setup, also in the European context. • Incorporate these algorithms in the verification and validation in the development process for their hardware components. • Potentially open new functionalities that can enable them to extend their hardware to a complete localization/perception functionality for automated vehicles

6. References

(1) *Robust Multi-Sensor Feature Matching for HD Map-Based Localization Using Graph Optimization*, P. Vieira Oliveira, R. Smit, A. Alassi, J. Brouwer, J. Goos, *IEEE ITSC, 2025*

(2) *SDS++: Online Situation-Aware Drivable Space Estimation for Automated Driving,* M. Muñoz Sánchez, G. Trots, R. Smit, P. Vieira Oliveira, J. Elfring, R. van de Molengraft, E. Silvas, *IEEE IV,* 2025